R PROGRAMMING – DATA VISUALIZATION ACTIVITY

Assessment: Scatterplots, Correlations, PCA, Network Graphs, Paired Charts, Bubble Charts and Hexbin Plots

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Aim

To perform data visualization and basic statistical analysis in R using different graphical techniques, including scatterplots, correlograms, PCA, network graphs, paired slope charts, bubble charts and hexbin plots.

Software / Requirements

• R and RStudio
• Required R packages: ggplot2, tidyr, igraph, Rtsne

General Procedure

1. Open RStudio and create a new R script.
2. Run the data-generation code given for each question.
3. Run the analysis and plotting code.
4. Record the numerical results shown in the Console.
5. Save the graphs from the Plots window where required.
6. Write the interpretation and recommendation below each result.

Q1 – Scatterplot

Study weekly sales (Rs lakh) against footfall (visitors per day) across 24 weekly records.

Tasks

Draw a scatterplot of sales against footfall with a straight-line fit and a loess curve.

Report the correlation and slope of the straight line and describe the strength and direction.

Identify the outlier using the largest standardized residual.

Fit a quadratic model and compare its R-squared with the straight-line model.

Give one recommendation for the Sales Director.

```
set.seed(1318053)
n <- 24
sales <- round(rnorm(n, 85.0, 18.0), 1)
z <- (sales - 85.0) / 18.0
footfall <- round(900 + 200 * (-0.70 * z + -0.71 * z^2 + 0.44 * rnorm(n)), 0)
footfall[19] <- round(footfall[19] + 3.9 * 200, 0)
df <- data.frame(sales, footfall)
```

```
library(ggplot2)

ggplot(df, aes(x = sales, y = footfall)) +
geom_point(size = 3) +
geom_smooth(method = "lm", se = FALSE) +
geom_smooth(method = "loess", se = FALSE, linetype = "dashed") +
labs(x = "Sales (Rs lakh)", y = "Footfall (visitors per day)",
title = "Weekly Sales vs Footfall") +
theme_minimal()

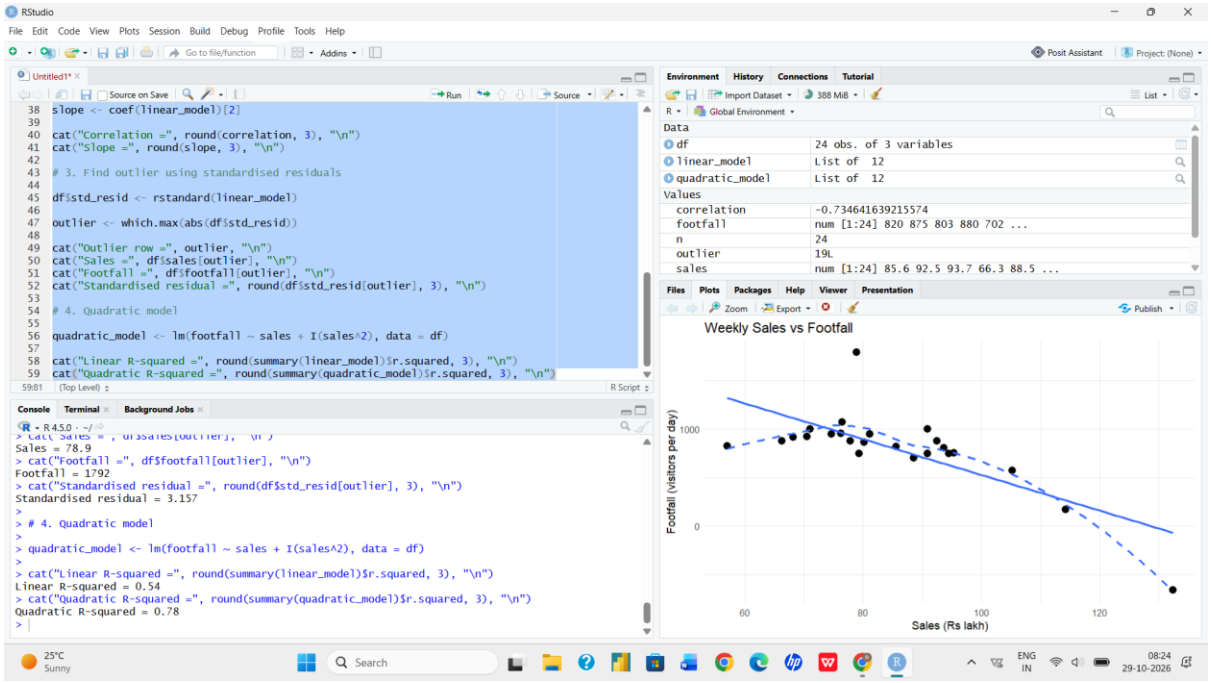
linear_model <- lm(footfall ~ sales, data = df)
correlation <- cor(df$sales, df$footfall)
slope <- coef(linear_model)[2]

df$std_resid <- rstandard(linear_model)
outlier <- which.max(abs(df$std_resid))

quadratic_model <- lm(footfall ~ sales + I(sales^2), data = df)

cat("Correlation =", round(correlation, 3), "\n")
cat("Slope =", round(slope, 3), "\n")
cat("Outlier row =", outlier, "\n")
cat("Sales =", df$sales[outlier], "\n")
cat("Footfall =", df$footfall[outlier], "\n")
cat("Linear R-squared =", round(summary(linear_model)$r.squared, 3), "\n")
cat("Quadratic R-squared =", round(summary(quadratic_model)$r.squared, 3), "\n")
```

Output / Result:



Interpretation: Compare the linear and quadratic R-squared values. The model with the higher R-squared gives the better fit.

Q2 – Correlogram

Study six measures recorded for 30 circle-periods at a telecom firm.

Tasks

Compute the correlation matrix, round it to two decimals, and draw a correlation heatmap.

Identify the two most strongly correlated variable pairs by absolute correlation.

Identify one surprising weak or negative correlation and explain what it implies.

Give one recommendation for the Network Head.

```
set.seed(1318106)
n <- 30
f1 <- rnorm(n)
f2 <- rnorm(n)

tower_density <- round(14.0 + 4.0 * (0.82 * f1 + 0.58 * rnorm(n)), 1)
drop_rate <- round(2.6 + 0.7 * (0.92 * f2 + 0.38 * rnorm(n)), 1)
traffic_tb <- round(3.2 + 0.32 * (0.58 * f2 + 0.20 * rnorm(n)), 1)
users_lakh <- round(24.0 + 6.0 * (-0.80 * f1 + 0.60 * rnorm(n)), 1)
churn_pct <- round(1.8 + 0.5 * (0.81 * f2 + 0.59 * rnorm(n)), 1)
arpu <- round(185 + 25.0 * (0.91 * f1 + 0.42 * rnorm(n)), 0)

df <- data.frame(tower_density, drop_rate, traffic_tb,
```

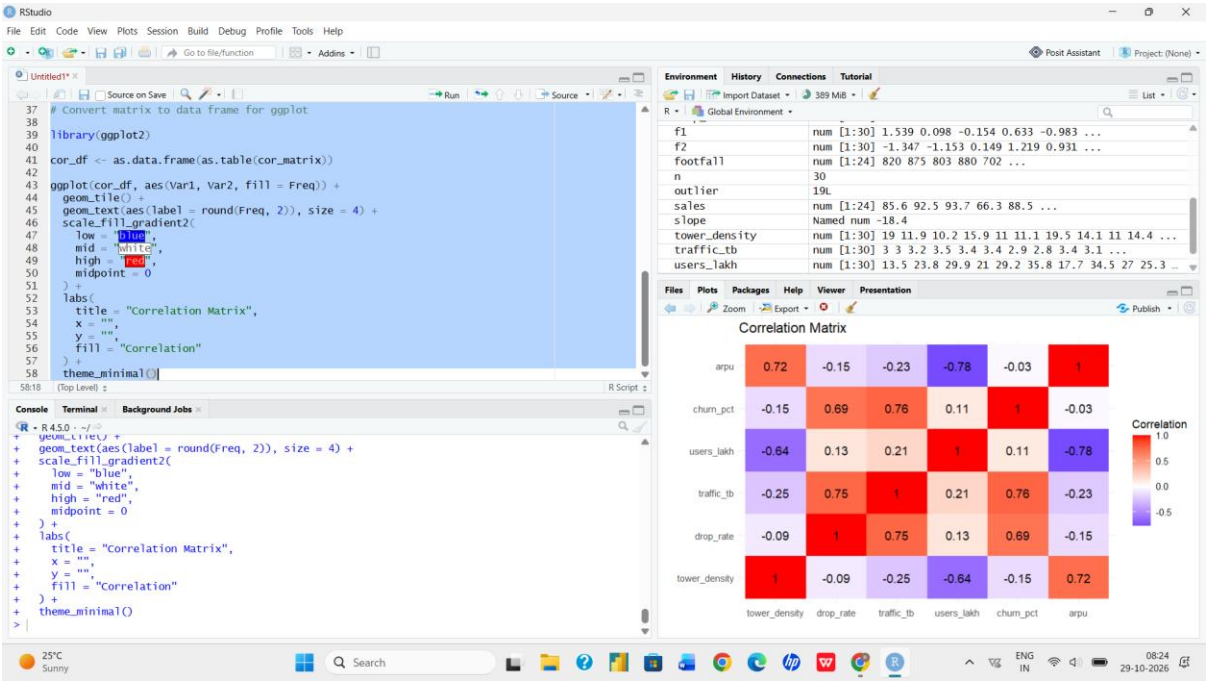
```
users_lakh, churn_pct, arpu)

cor_matrix <- cor(df)
print(round(cor_matrix, 2))

library(ggplot2)
cor_df <- as.data.frame(as.table(cor_matrix))

ggplot(cor_df, aes(Var1, Var2, fill = Freq)) +
geom_tile() +
geom_text(aes(label = round(Freq, 2)), size = 4) +
scale_fill_gradient2(low = "blue", mid = "white",
high = "red", midpoint = 0) +
labs(title = "Correlation Matrix", x = "", y = "", fill = "Correlation") +
theme_minimal()
```

Output / Result:



Q3 – Dimension Reduction (PCA and t-SNE)

Six measures were recorded for 36 destinations from three known groups.

Tasks

Perform PCA on scaled data. Report the percentage of variance for PC1 and PC2 and the number of components needed for at least 80% variance.

Plot PC1 against PC2 coloured by group and report the variable with the largest absolute loading on PC1.

Judge how well the groups separate using the share of variance in each PC and the R-squared of lm(score ~ grp).

Make a t-SNE plot and explain the variance-versus-neighbourhood trade-off between PCA and t-SNE.

```
set.seed(1318182)
grp <- rep(c("Group C", "Group B", "Group A"), each = 12)
g <- rep(c(-1, 0, 1), each = 12)

footfall_k <- round(120 + 30 * (rnorm(36) + 0.90 * g), 1)
spend_rs <- round(3200 + 600 * (rnorm(36) + 1.10 * g), 1)
stay_nights <- round(2.8 + 0.7 * (rnorm(36) + 0.70 * g), 1)
rating10 <- round(7.6 + 0.8 * (rnorm(36) + 1.00 * g), 1)
occupancy_pct <- round(68 + 10 * (rnorm(36) + 0.06 * g), 1)
complaints <- round(4.0 + 1.2 * (rnorm(36) + 1.20 * g), 1)

df <- data.frame(footfall_k, spend_rs, stay_nights,
rating10, occupancy_pct, complaints)

pca <- prcomp(df, scale. = TRUE)
variance <- pca$sdev^2 / sum(pca$sdev^2)
cumulative <- cumsum(variance)

cat("Variance PC1 (%) =", round(variance[1] * 100, 2), "\n")
cat("Variance PC2 (%) =", round(variance[2] * 100, 2), "\n")
cat("Components for 80% =", which(cumulative >= 0.80)[1], "\n")
print(round(pca$rotation, 3))

scores <- as.data.frame(pca$x)
scores$grp <- grp

library(ggplot2)
ggplot(scores, aes(PC1, PC2, colour = grp)) +
geom_point(size = 3) +
labs(title = "PCA: PC1 vs PC2") +
theme_minimal()

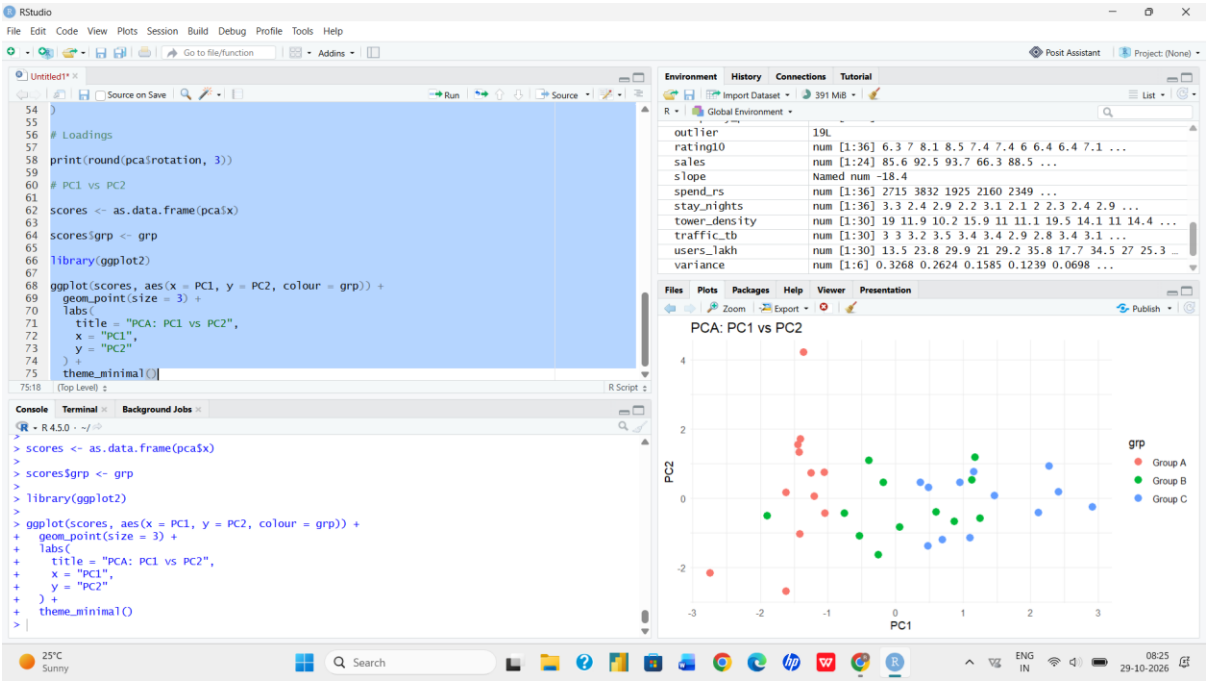
group_model <- lm(PC1 ~ grp, data = scores)
cat("Group R-squared =", round(summary(group_model)$r.squared, 3), "\n")

library(Rtsne)
set.seed(100)
tsne_result <- Rtsne(scale(df), dims = 2, perplexity = 8,
check_duplicates = FALSE)
```

```
tsne_df <- data.frame(
X = tsne_result$Y[,1],
Y = tsne_result$Y[,2],
grp = grp
)

ggplot(tsne_df, aes(X, Y, colour = grp)) +
geom_point(size = 3) +
labs(title = "t-SNE Plot") +
theme_minimal()
```

Output / Result:



Interpretation: PCA preserves overall variance and is easier to interpret globally, while t-SNE focuses more on local neighbourhood relationships.

Q4 – Network Graph

Study parcel-delivery connections between eight hubs. Each row represents a link with a weight.

Tasks

Build and draw the network with node size proportional to degree and edge width proportional to weight.

Compute degree and betweenness for every node and identify the most central node.

Report the number of edges, network density and number of connected components.

Delete the most central node and report the number of components remaining. Give one recommendation.

```
set.seed(1318212)

nodes <- c("Mumbai", "Ahmedabad", "Delhi", "Chennai",
"Kolkata", "Hyderabad", "Bengaluru", "Pune")

pairs <- t(combn(nodes, 2))
u <- runif(nrow(pairs))

keep <- (u < 0.30) |
((pairs[,1] == "Kolkata" | pairs[,2] == "Kolkata") & u < 0.93)

edges <- data.frame(
from = pairs[keep, 1],
to = pairs[keep, 2],
weight = round(runif(sum(keep), 1, 20))
)

library(igraph)

g <- graph_from_data_frame(edges, directed = FALSE, vertices = nodes)

V(g)$size <- degree(g) * 5 + 10
E(g)$width <- E(g)$weight / 3

plot(g, vertex.label = V(g)$name)

degree_values <- degree(g)
betweenness_values <- betweenness(g)

network_stats <- data.frame(
Node = names(degree_values),
Degree = degree_values,
Betweenness = round(betweenness_values, 2)
)

print(network_stats)

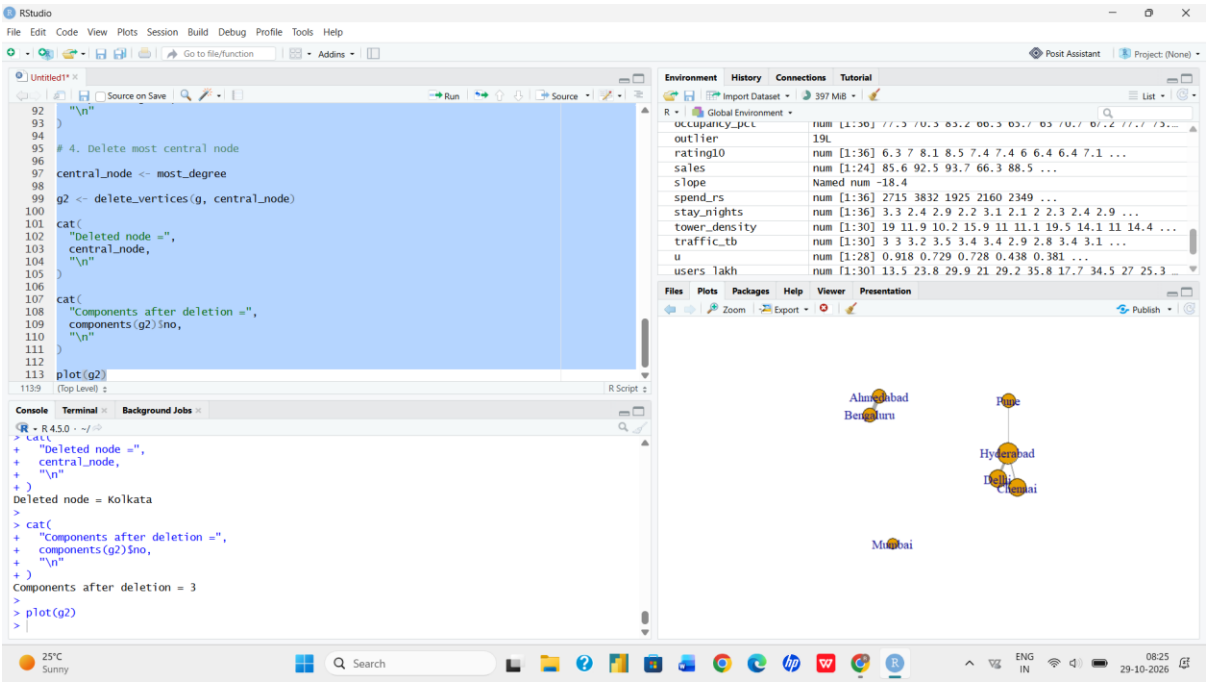
cat("Number of edges =", ecount(g), "\n")
cat("Network density =", round(edge_density(g), 3), "\n")
cat("Connected components =", components(g)$no, "\n")
```

```
central_node <- names(which.max(degree_values))
g2 <- delete_vertices(g, central_node)

cat("Deleted node =", central_node, "\n")
cat("Components after deletion =", components(g2)$no, "\n")

plot(g2)
```

Output / Result:



Q5 – Paired / Slope Chart

Staff-on-duty values were recorded before and after a process change at six departments.

Tasks

Draw a paired slope chart with two points per entity joined by a line and labelled endpoints.

Classify each entity as improved, worsened or flat and report the counts.

Quantify the largest change and its percentage of the before value.

Run a paired t-test and report t, p and mean change. Give one recommendation.

```
set.seed(1318294)

entity <- c("Radiology", "Pharmacy", "Orthopaedics",
"Cardiology", "Emergency", "Oncology")

n <- 6
before <- round(rnorm(n, 18, 2.0), 0)
```

```
change <- round(
c(-1, -1, 1, -1, -1, -1) *
runif(n, 0.4, 1.4) * 2.4, 0
)

change[2] <- 0.0
after <- before + change

df <- data.frame(entity, before, after)
print(df)

library(ggplot2)
library(tidyr)

long_df <- pivot_longer(
df, cols = c(before, after),
names_to = "time", values_to = "staff"
)

ggplot(long_df, aes(time, staff, group = entity)) +
geom_line() +
geom_point(size = 3) +
geom_text(aes(label = staff), vjust = -0.8) +
labs(title = "Staff Before and After Process Change",
x = "", y = "Staff on Duty") +
theme_minimal()

df$change_value <- df$after - df$before

df$status <- ifelse(
abs(df$change_value) < 0.18, "Flat",
ifelse(df$change_value > 0, "Improved", "Worsened")
)

print(df)

largest <- which.max(abs(df$change_value))

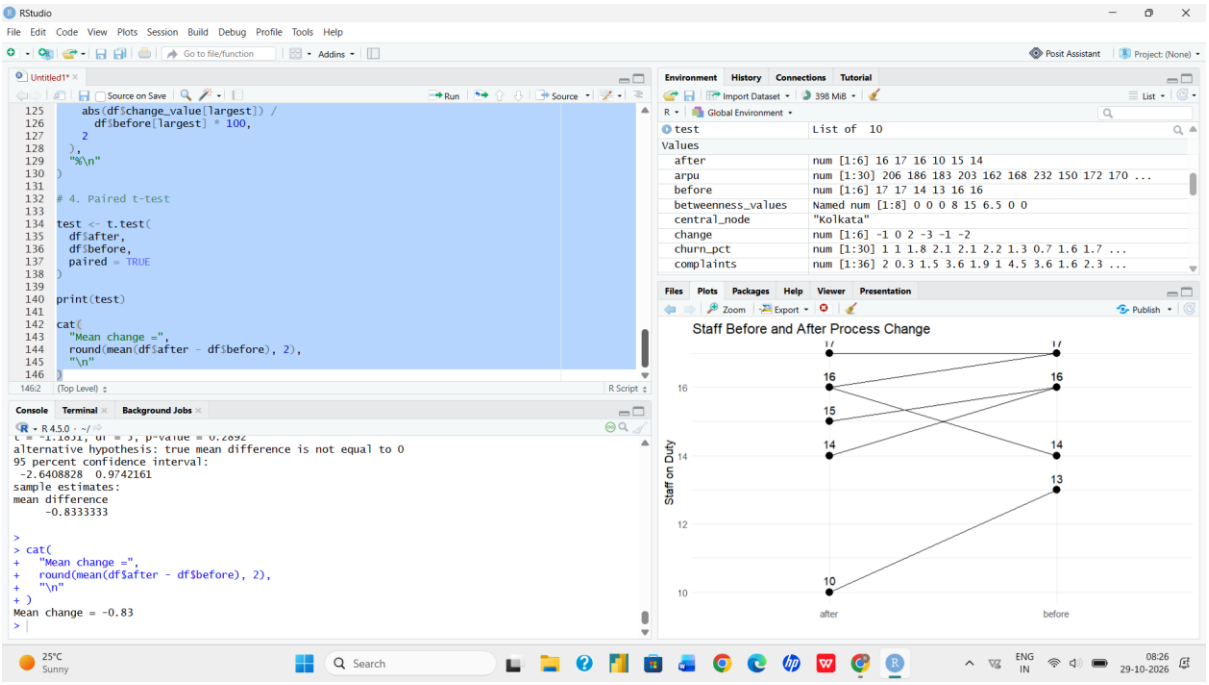
cat("Improved =", sum(df$status == "Improved"), "\n")
cat("Worsened =", sum(df$status == "Worsened"), "\n")
cat("Flat =", sum(df$status == "Flat"), "\n")
cat("Largest change =", df$entity[largest], "\n")
cat("Change =", df$change_value[largest], "\n")
cat("Percentage =", round(abs(df$change_value[largest]) /
```

df$before[largest] * 100, 2), "%\n")

test <- t.test(df$after, df$before, paired = TRUE)
print(test)

cat("Mean change =", round(mean(df$after - df$before), 2), "\n")

Output / Result:



Q6 – Bubble Chart

Compare genres using marketing spend, watch time and subscriber count.

Tasks

Draw a bubble chart with marketing spend on x, watch time on y and subscribers as bubble size. Label every bubble.

Report the correlation between x and y.

Identify the subscriber-size outlier using a z-score.

State how many times larger its size is than a typical non-outlier and explain why a plain scatterplot could hide it.

Give one recommendation for the Content Head.

set.seed(1318318)

entity <- c("Sci-Fi", "Romance", "Documentary", "Sports",
"Drama", "Thriller", "Animation")

```
n <- 7
z <- rnorm(n)

spend_lakh <- round(45.0 + 12.0 * z, 1)
watch_hrs <- round(11.0 + 0.90 * z + 0.40 * rnorm(n), 1)
subs_lakh <- round(18.0 + 4.0 * rnorm(n), 1)

subs_lakh[4] <- round(18.0 * 2.4, 1)
spend_lakh[4] <- round(45.0)
watch_hrs[4] <- round(11.0)

df <- data.frame(entity, spend_lakh, watch_hrs, subs_lakh)
print(df)

library(ggplot2)

ggplot(df, aes(spend_lakh, watch_hrs, size = subs_lakh)) +
geom_point(alpha = 0.7) +
geom_text(aes(label = entity), vjust = -1) +
scale_size_area() +
labs(title = "Marketing Spend, Watch Time and Subscribers",
x = "Marketing Spend (Rs lakh)",
y = "Watch Time (hours/user/month)",
size = "Subscribers (lakh)") +
theme_minimal()

cat("Correlation =", round(cor(df$spend_lakh, df$watch_hrs), 3), "\n")

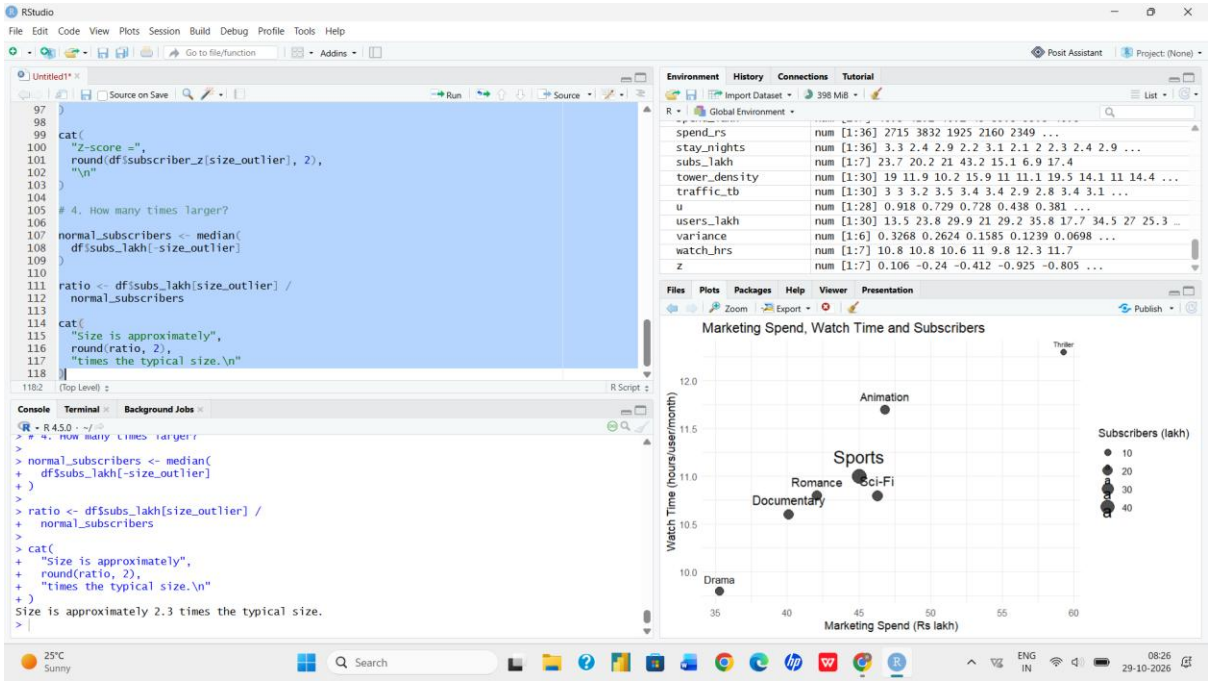
df$subscriber_z <- as.numeric(scale(df$subs_lakh))
size_outlier <- which.max(abs(df$subscriber_z))

cat("Size outlier =", df$entity[size_outlier], "\n")
cat("Z-score =", round(df$subscriber_z[size_outlier], 2), "\n")

normal_subscribers <- median(df$subs_lakh[-size_outlier])
ratio <- df$subs_lakh[size_outlier] / normal_subscribers

cat("Size is approximately", round(ratio, 2),
"times the typical size.\n")
```

Output / Result:



Q7 – Hexbin Plot

Study loan amount and applicant income for 3,000 records.

Tasks

Draw a hexbin plot with a count legend and a plain scatterplot.

Using a 10 × 10 grid, find the busiest cell and report its x range, y range, count and share of all points.

Report the number of distinct (x, y) pairs and explain why the scatterplot alone may be misleading.

Give one recommendation for the Zonal Manager.

```
set.seed(1318408)

n <- 3000
core <- runif(n) < 0.85

loan_lakh <- round(
ifelse(core, rnorm(n, 38.0, 5.4),
runif(n, 11.0, 65.0)), 1
)

income_k <- round(
ifelse(core,
52.0 - 0.30 * (loan_lakh - 38.0) / 5.4 +
```

```
sqrt(1 - 0.38) * 7.2 * rnorm(n),
runif(n, 16.0, 88.0)), 1
)

df <- data.frame(loan_lakh, income_k)

library(ggplot2)

ggplot(df, aes(loan_lakh, income_k)) +
geom_hex(bins = 25) +
labs(title = "Loan Amount vs Applicant Income",
x = "Loan Amount (Rs lakh)",
y = "Applicant Income (Rs thousand/month)",
fill = "Count") +
theme_minimal()

ggplot(df, aes(loan_lakh, income_k)) +
geom_point(alpha = 0.3) +
labs(title = "Plain Scatterplot",
x = "Loan Amount (Rs lakh)",
y = "Applicant Income (Rs thousand/month)") +
theme_minimal()

x_bins <- cut(df$loan_lakh, breaks = 10, include.lowest = TRUE)
y_bins <- cut(df$income_k, breaks = 10, include.lowest = TRUE)

grid <- table(x_bins, y_bins)
max_cell <- which(grid == max(grid), arr.ind = TRUE)

print(grid)

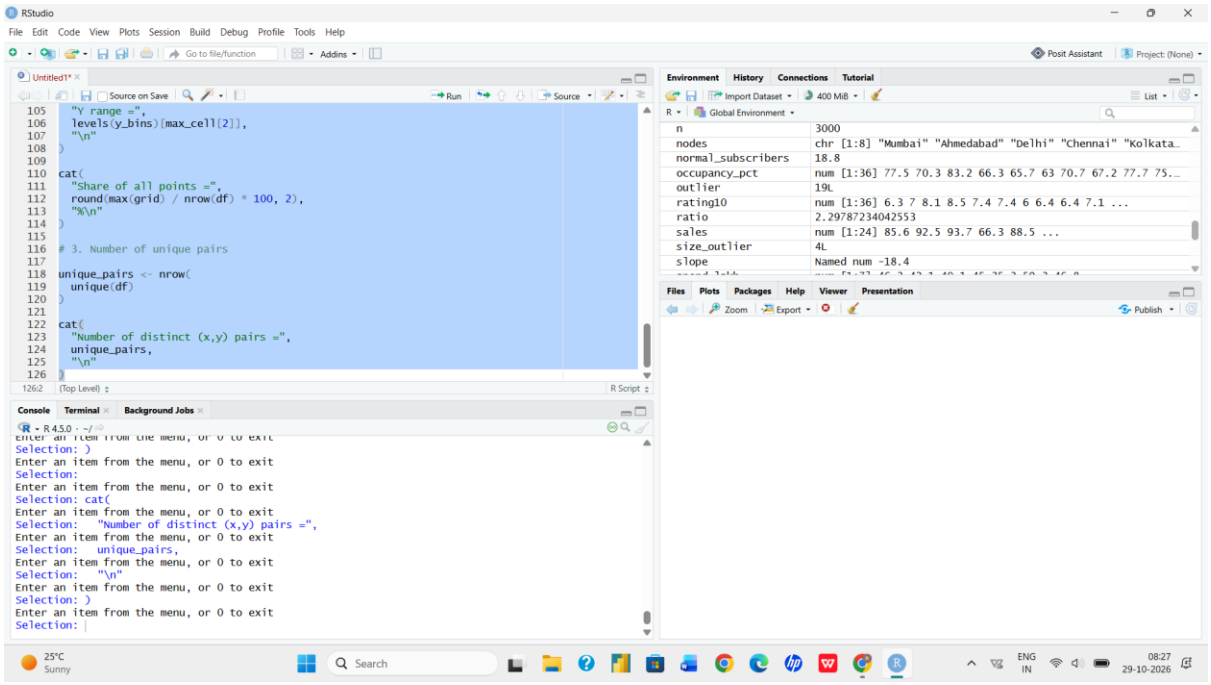
cat("Busiest cell count =", max(grid), "\n")
cat("X range =", levels(x_bins)[max_cell[1]], "\n")
cat("Y range =", levels(y_bins)[max_cell[2]], "\n")
cat("Share of all points =", round(max(grid) / nrow(df) * 100, 2), "%\n")

unique_pairs <- nrow(unique(df))

cat("Distinct (x,y) pairs =", unique_pairs, "\n")
```

Output / Result:
Insert the hexbin and plain scatterplot here. Record the busiest cell ranges, count, percentage share and distinct (x,y) pairs.

Interpretation: A scatterplot can hide overlapping observations. A hexbin plot is more useful for this large dataset because it shows the concentration of observations in different regions.



Overall Result

The activity demonstrates the use of R for statistical visualization and exploratory data analysis. Different plots are used according to the structure of the data: scatterplots for relationships, correlograms for correlations, PCA and t-SNE for dimension reduction, network graphs for connections, slope charts for before-after comparisons, bubble charts for three-variable comparisons, and hexbin plots for dense datasets.

Conclusion

Thus, the given datasets were analysed using R and appropriate visualization techniques. The graphs, statistical measures and interpretations help in identifying relationships, outliers, clusters, central nodes and high-density regions for decision making.